QUBO Formulation: Quantum Circuit Compilation Candidate Selection Problem

Quantum Circuit Compilation Candidate Selection Problem

Problem Statement. Given a set of N candidate compiled circuits, the objective is to select exactly one candidate that minimizes the total compilation cost. Each candidate $i \in \{0, \ldots, N-1\}$ is associated with a pre-computed scalar cost c_i, which aggregates depth, fidelity loss, and hardware constraint penalties. The problem requires choosing a single candidate such that the sum of the costs of the selected candidates is minimized, subject to the constraint that precisely one candidate is chosen.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each candidate $i \in \{0, \ldots, N-1\}$. The variable x_i takes the value 1 if candidate i is selected, and 0 otherwise.

QUBO Derivation. We formulate the problem as a Quadratic Unconstrained Binary Optimization (QUBO) by converting the constrained minimization into an unconstrained energy minimization.

Step 1 — Objective Function. The original objective is to minimize the total cost:

$$\min_x \; C(x) = \sum_{i=0}^{N-1} c_i x_i. \tag{1}$$

Since QUBO minimization is equivalent to the original minimization, we retain this form.

Step 2 — Constraint. The requirement to select exactly one candidate is expressed as:

$$\sum_{i=0}^{N-1} x_i = 1. \tag{2}$$

Step 3 — Penalty Term. We enforce the constraint by adding a quadratic penalty term $P(x)$ with weight $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2. \tag{3}$$

Expanding the square yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right). \tag{4}$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, we simplify:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right) \tag{5}$$

$$= \lambda \left(- \sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j + 1 \right) \tag{6}$$

$$= -\lambda \sum_{i=0}^{N-1} x_i + \lambda \sum_{i \neq j} x_i x_j + \lambda. \tag{7}$$

Step 4 — Combined Hamiltonian. The total QUBO Hamiltonian $H(x)$ is the sum of the objective and penalty terms:

$$H(x) = C(x) + P(x) \tag{8}$$

$$= \sum_{i=0}^{N-1} c_i x_i - \lambda \sum_{i=0}^{N-1} x_i + \lambda \sum_{i \neq j} x_i x_j + \lambda \tag{9}$$

$$= \sum_{i=0}^{N-1} (c_i - \lambda) x_i + \lambda \sum_{i \neq j} x_i x_j + \lambda. \tag{10}$$

Step 5 — Q Matrix and Offset. The Hamiltonian is expressed in the standard QUBO form $H(x) = \sum_{i \leq j} Q_{ij} x_i x_j + c$. Comparing terms, the matrix entries Q_{ij} and offset c are given by:

$$Q_{ii} = c_i - \lambda, \quad \forall i \in \{0, \ldots, N-1\}, \tag{11}$$

$$Q_{ij} = \lambda, \quad \forall i, j \in \{0, \ldots, N-1\},\ i \neq j, \tag{12}$$

$$c = \lambda. \tag{13}$$

Note that the term $\lambda \sum_{i \neq j} x_i x_j$ corresponds to symmetric matrix entries $Q_{ij} = \lambda$ for $i \neq j$.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any infeasible solution has strictly higher energy than any feasible solution. A sufficient condition is:

$$\lambda > \max_{i \in \{0, \ldots, N-1\}} c_i. \tag{14}$$

This choice guarantees that the minimum penalty for a constraint violation (which occurs when zero candidates are selected, yielding penalty λ) exceeds the cost of any single candidate. For violations involving multiple selections ($k \geq 2$), the penalty grows quadratically as $\lambda(k-1)^2 \geq \lambda$, further increasing the energy. Thus, $\lambda > \max_i c_i$ ensures that the global minimum of $H(x)$ cannot occur at an infeasible point.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal compiled circuit candidate. For any feasible solution x (where $\sum_i x_i = 1$), the penalty term vanishes, and $H(x) = C(x)$. For any infeasible solution, $H(x) = C(x) + P(x)$. Under the condition $\lambda > \max_i c_i$, the energy of any infeasible solution exceeds $\max_i c_i$, while the energy of any feasible solution is at most $\max_i c_i$. Consequently, the global minimum must be feasible. Among feasible solutions, $H(x)$ equals the objective $C(x)$, so the minimizer of $H(x)$ is the candidate with the minimum compilation cost.

Illustrative Example. Consider an instance with $N = 3$ candidates and costs $c_0 = 10$, $c_1 = 20$, $c_2 = 15$. We choose $\lambda = 21$, satisfying $\lambda > \max\{10, 20, 15\}$.

The concrete QUBO parameters are:

$$Q_{00} = 10 - 21 = -11, \tag{15}$$
$$Q_{11} = 20 - 21 = -1, \tag{16}$$
$$Q_{22} = 15 - 21 = -6, \tag{17}$$
$$Q_{ij} = 21 \quad \text{for } i \neq j, \tag{18}$$
$$c = 21. \tag{19}$$

The Hamiltonian is:

$$H(x) = -11x_0 - x_1 - 6x_2 + 21(x_0x_1 + x_0x_2 + x_1x_2) + 21. \tag{20}$$

Evaluating $H(x)$ for candidate solutions:

- $x = [1, 0, 0]$: $H = -11 + 21 = 10$.
- $x = [0, 1, 0]$: $H = -1 + 21 = 20$.
- $x = [0, 0, 1]$: $H = -6 + 21 = 15$.
- $x = [0, 0, 0]$: $H = 21$.
- $x = [1, 1, 0]$: $H = -11 - 1 + 21 + 21 = 30$.

The global minimum is $H = 10$ at $x = [1, 0, 0]$, corresponding to selecting candidate 0 with minimum cost 10.